Quality Plan

Quality Plan: ShopForMe

Project Title: ShopForMe

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Version: 1.1

Summary

This document defines the general framework of the project and the management and quality practices applied to the project.

The project quality plan is a document structured under five headings:

1. *Purpose of the Project Quality Plan*
2. *General information about the project*
3. *Project organisation*
4. *Development plan*
5. *Quality assurance*

I - Purpose of the Project Quality Plan

I.1 - Project Quality Plan framework

The aim of the Project Quality Plan is to define and monitor the measures to be taken as part of the ***ShopForMe*** project in order to ensure quality and achieve the expected results.

To this end, the Quality Plan will set out the rights and duties of each stakeholder with a view to guaranteeing this objective. More specifically, the following will be determined by mutual agreement

- The overall organisation of the project in terms of the structures to be implemented.
- The detailed project development plan.
- The project management plan.
- The limits of the project.

I.2 - Use of the Project Quality Plan

A monitoring and management tool, it is used throughout the project and incorporates new data. It is therefore intended to be constantly adjusted according to circumstances, in line with a corrective action procedure defined in this document.

At the end of the project, the modified and annotated Quality Plan will be one of the project's outcome documents.

II - General information about the project

II.1 - Definition of quality objectives

- **Context** : The ***ShopForMe*** project aims to improve the lives of visually impaired people by offering them new possibilities in terms of what they can and cannot do.
  People are very attached to their lifestyle and, unfortunately, accidents can happen to anyone at any time. This type of event can drastically alter people's daily lives, making it impossible for them to carry out certain activities, such as shopping, for example, which represents the most ordinary outing in our society.
  The aim is to provide visually impaired people with a tool that will enable them to continue this type of activity and carry out their shopping independently, without the need for a third party.

- **Goals** : The project must follow a precise path and comply with a number of criteria that are essential for the application to work properly and for user safety:
  - Ensure safe and precise navigation.
  - Navigation must be based on the shortest route.

II.2 - Quality requirements

- **Functional** :
  - Accurate user location.
  - Clear voice and visual guidance.
  - Accessible user interface.
  - Product search function.

- **Non-functional** :
  - Performance : Fast response from the application, with less than 2 seconds latency for guidance actions.
  - Location : Accuracy of location to be equivalent to a margin of error of 1 metre.
  - Security : Protection of users' personal data.
  - Reliability : 99.9% system availability.
  - Accessibility : Compliance with WCAG 2.1 standards (level AA).

II.3 - Test Strategy

II.3.1 - General tests

- **Types of tests** :

- Unit Testing : Checking the completeness of the source code: every method and condition must be tested.
- Integration tests : Validation of the interaction between the various parties.
- Acceptance tests : Based on real scenarios and validated by end users.
- Performance tests : Measurement of different response times.
- IoT testing : Specific validation of hardware components, including ESP32.

- **Validation methods** :
  - Test coverage : Objective of at least 85% overall coverage on the project.
  - Using Sonar : Guaranteeing high-quality, optimised code.
  - Pipeline : The pushed code must pass all the tests.

II.3.2- IoT Tests

- **Connectivity Tests** :
  - Checks the stability of the ESP32's Wi-Fi connection.
  - Bluetooth connectivity tests to ensure seamless communication with mobile devices.

- **ESP32 Performance Tests :**
  - Measurement of ESP32 response time for guidance actions.
  - Load tests to assess performance under different conditions of use.

- **Durability and Reliability Tests :**
  - Simulation of prolonged use to assess the durability of the ESP32.
  - Tests for resistance to electromagnetic interference.

- **Safety tests :**
  - Verification of security measures to protect communications between the ESP32 and other system components.

- **IoT Integration Tests :**
  - Validation of the interaction between the ESP32 and the other components of the system (mobile application, backend servers).

- **Energy Consumption Tests :**
  - Measurement of ESP32 energy consumption to optimise autonomy.

II.4 - Project deliverables

The ‘Project Deliverables’ section is crucial for clearly defining what will be produced during the project. The different outputs and requirements for each party must be highlighted.

II.4.1 - Documentation

- **Functional Specifications** : Document detailing the functionality of the application, including user requirements and use cases.

- **Technical Specifications** : Contains the complete system architecture, UML diagrams, implementation details and API specifications.

- **User manual** : Step-by-step instructions for installing and configuring the application.

- **Test plan** : Description of test strategy and objectives :
  - Test case : A set of preconditions, input data, actions and expected results, developed to verify the correct operation of a development or product.
  - Automated test scripts : Set of instructions for automatically testing an application.

II.4.2 - Software

- **Source code** : Complete source code for the application, well commented and structured, with instructions for deployment.

- **Installation and Deployment Scripts** : Scripts to automate the installation and updating of the application.

II.4.3 - Tests

- **Test results** : detailed results of all tests carried out, including unit, integration, acceptance and accessibility tests.

- **Bug and Patch Reports** : List of bugs detected, their severity, their status and the fixes applied.

II.4.4 - Deployment and Maintenance

- **Deployment Plan** : Deployment strategy, including stages, resources required, timetable and validation procedures.

- **Maintenance Plan** : Post-deployment maintenance strategy, including management of updates, security patches and ongoing support.

III - Project organisation

III.1 - **Lists of project participants**

- **Team** :
  - Project manager : Responsible for general coordination.
  - Developers: responsible for programming and implementation.
  - QA Tester: Responsible for testing and validation.
  - UX/UI Designer: Ensures the interface is accessible and easy to use.

- **Tools** :
  - Project management : GitHub.

- Development : Visual Studio, Android Studio.
- Tests : JMeter, Mockito, Appium, Cypress.
- CI/CD : Jenkins, GitHub.
- Code analysis : SonarQube
- Security: OWASP ZAP

III.2 - Procedure for validating project documents

In order to limit delays linked to the validation of project documents, any delivery or update of a project document will be carried out as follows:

- **Delivery notification:** The writer will notify the recipient(s) of the delivery on date t.

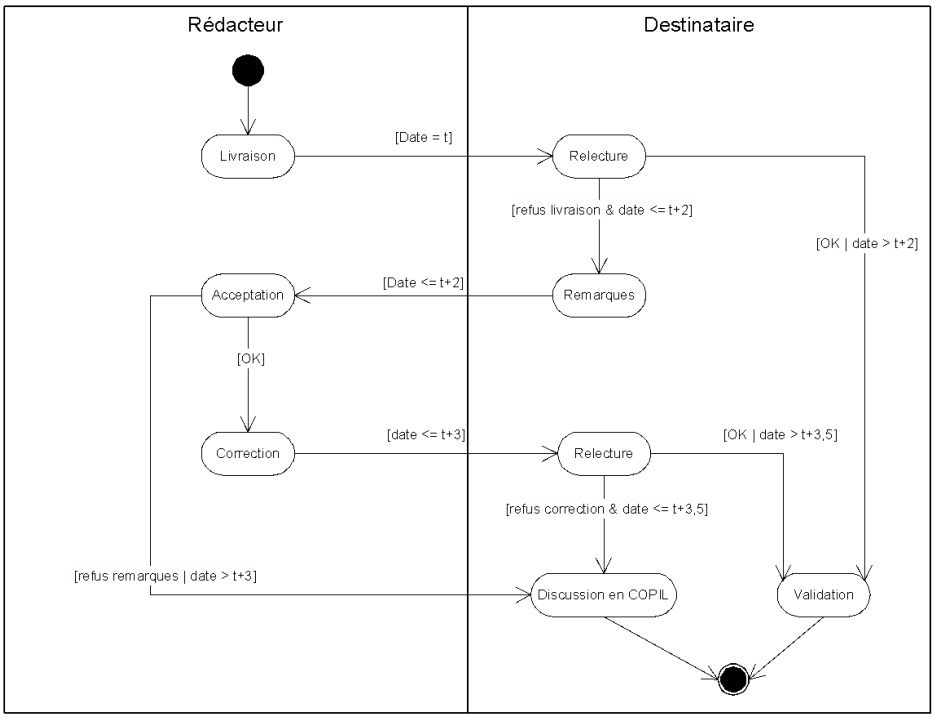
- **Initial review:** From the date of delivery, the recipient has 2 working days to report any comments (or lack of comments) to the writer.

- **Correction and Revision:**
  - From the time the remarks are received, the editor has one working day to correct or contest the remarks and deliver the correction.
  - Once the correction has been delivered, the recipient has half a working day to validate the corrections.

- **Final validation :**
  - Validation of the document is discussed by the steering committee in the event of :
    - The writer exceeds the deadline.
    - Comments are rejected.
    - Corrections are rejected.

  - Validation is effective in the event of :
    - The customer agrees.
    - Deadline exceeded by the recipient.

By following these procedures, we ensure that project documents are validated efficiently and on time, minimising delays and misunderstandings.



IV - Development plan

IV.1 - Project success factors and risks

Factors that could influence or impact the project, either positively or negatively :

- Description of success factors :
  - Involvement of future users (visually impaired).
  - Clear objectives and specifications.
  - Short deadlines.
  - Realistic expectations.

- Identification of major project risks:
  - Lack of information about the end user.
  - Poor estimation of workloads.
  - Inadequate project monitoring.
  - Uncontrolled modifications.

- Preventive action to be incorporated into the development plan:
  - Check schedules and measure progress regularly.

IV.2 - Indicators

Indicators of success, project monitoring, risk monitoring, etc.

- Success indicators :

    - Number of anomalies detected.
    - Number of resolved anomalies to be verified.

- Project monitoring indicator :
    - Percentage of functions analysed.
    - Percentage of functions developed.
    - **For each function:** Percentage of estimated actual progress.

- Risk monitoring indicator:
    - **For each function:** Percentage of tests that cannot be carried out.
    - **For each function:** Percentage of compliant tests.

V - Quality assurance

- **Introduction to quality assurance:** Quality assurance is a set of systematic processes that guarantee that the products or services provided by the project meet specific requirements and are reliable. It is essential for maintaining customer satisfaction and ensuring the delivery of a high quality product.

- **Standards and procedures:** Standards may include internal company standards, industry standards or international standards such as ISO 9001. Procedures describe how these standards will be implemented within the project. For example, you may have a procedure for code review or bug management.

- **Responsibilities:** It's important to define who in the team is responsible for quality assurance. This may be a dedicated role, such as a QA engineer, or it may be a responsibility shared by the whole team.

- **Quality assurance methods:** Quality assurance methods can include code reviews, unit tests, integration tests, performance tests, quality audits, etc. It is important to describe how these methods are used. It is important to describe how these methods will be used in the project.

- **Quality monitoring plan:** This describes how quality will be monitored and measured throughout the project. This may include specific metrics, such as the number of bugs per release, or the percentage of successful tests.

- **Managing quality issues:** When a quality issue is identified, it is important to have a process for managing it. This can include documenting the problem, determining its cause, implementing a solution, and checking that the solution has solved the problem.

- **Continuous improvement:** Quality assurance is not a one-off process, but an ongoing effort to improve quality. This can include using customer feedback, analysing past quality problems to prevent them recurring, and constantly looking for new ways to improve quality.